FFGFT — Information and Black Holes

What Is Fundamental Information? Ontology vs. Epistemology

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https://github.com/jpascher/T0-Time-Mass-Duality

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Abstract

This document clarifies what *information* means in FFGFT in the context of black holes. It shows that the Hawking information paradox rests on a category confusion: the classical formulation treats information as an epistemic concept. FFGFT distinguishes the **fundamental information**: the topological structure of winding numbers on T^4 — comparable to a topological charge, not to a quantity of bits. This topological charge is **topologically preserved**, independently of any observer. The compactification of the time dimension on T^4 undermines one of the three presuppositions of the paradox, and thermodynamic irreversibility is fully compatible with topological information preservation.

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.1 The Hawking Information Paradox and Its Presuppositions

Stephen Hawking showed in 1974 that black holes emit thermal radiation [**?**]. This radiation carries no information about the specific state of the infalling matter. If the black hole evaporates completely, the original quantum information would be destroyed — violating unitarity.

The paradox presupposes three conditions that do not all hold in FFGFT:

1. Information is the complete description of the quantum state of all infalling particles.
2. Time is continuous and unidirectionally oriented — what fell into the black hole "in the past" is "forever" inaccessible.
3. The classical singularity is physically real — the state of matter is destroyed there.

Important: [Why does the paradox arise at all?] Hawking (implicitly) presupposes that information is *epistemic* — what an observer can reconstruct — and that time is continuous. Neither is fundamental in FFGFT: information is a topological charge, not an epistemic quantity; and time is compactified on T^4, not continuous. The paradox dissolves not by adding new physics, but by clarifying the presuppositions.

FFGFT rejects all three presuppositions in specific ways — not by assertion, but through the geometric structure of the framework.

.2 What Information in FFGFT Is *Not*

Not our description of the black hole

Information in the FFGFT sense is **not** our knowledge or understanding of the black hole. The Hawking paradox is an **epistemic** problem: it asks whether an observer outside the horizon can fully reconstruct the original quantum state. FFGFT answers a different question: does the fundamental topological structure continue to exist?

Important: Information in the sense of FFGFT is not an epistemic category (what we can know), but an **ontological** category (what structurally exists). The inability of an external observer to reconstruct information is an access problem — not an existence problem.

Observable consequence: If the topological information were truly destroyed, the Hawking radiation would be exactly black-body radiation. The modified spectrum (due to the T^4 winding structure) is the falsifiable difference — this empirically grounds the access problem.

Not the sum of all detail states

Information in the FFGFT sense is also not the sum of all information required to fully describe all particles. These detail states can be epistemically lost through thermalisation. That is thermodynamic irreversibility, not ontological information destruction.

.3 What Information in FFGFT *Is*: Fundamental Information

Topological invariants as fundamental information

In FFGFT, particles are winding configurations on the 4D identification torus T^4 with quantum numbers $(n_\theta, n_\varphi, n_3, n_4)$. These winding numbers are **topological invariants**.

Definition of Fundamental Information

The fundamental information of a physical system in FFGFT is the **topological charge** of its constituents: the winding numbers $(n_\theta, n_\varphi, n_3, n_4)$ on T^4. These define *what kind* of particles exist — not *where* they are, not *how* they behave, not *in what exact quantum state* they find themselves.
Fundamental information is not a *quantity of bits* (Shannon), but a **topological charge** — comparable to a magnetic monopole charge or a winding number in string theory. This immediately decouples FFGFT from the usual bit-oriented information paradox debate.

Topological invariants cannot be reduced to zero through continuous transformations. A state with winding number $n = 1$ cannot be continuously deformed into $n = 0$ without a topological discontinuity — with measurable physical consequences.

No destruction operator acts on topological invariants. This is the central statement: there is no physical process that can reduce winding numbers to zero without generating a measurable discontinuity. Classical singularities would be such a process — which is why they are structurally excluded in FFGFT.

The foundational relation as structure preservation

The fundamental relation $\tilde{T} \cdot m = 1$ connects the T-field and mass at every point. In the limit of extreme compression:

$$\tilde{T} \to L_0/c = \xi \cdot t_P > 0 \tag{1}$$

The T-field never reaches zero — the minimum value is L_0/c. The winding structure of the field configuration is preserved.

Algebraic protection against singularities (Dok. 078)

Here lies the most elegant proof — not by a postulate, but by the algebraic structure of the foundational relation itself.

From $\tilde{T} \cdot m = 1$:

$$\tilde{T} = \frac{1}{m} \tag{2}$$

If $m \to \infty$, then $\tilde{T} \to 0$. But reaching $\tilde{T} = 0$ exactly would require $m = \infty$ exactly — algebraically impossible.

Algebraic Protection — Dok. 078

$\tilde{T} \cdot m = 1$ acts as algebraic protection against singularities:

- $m \to \infty$ implies $\tilde{T} \to 0$ — but only **asymptotically**, never exactly.
- $\tilde{T} = 0$ exactly would require $m = \infty$ exactly — algebraically impossible.
- The singularity is not a forbidden state, but an **unreachable limit**.

This is not a geometric and not a quantum mechanical argument — it is an **algebraic argument** from the foundational relation alone. It holds independently of the size of the black hole and regardless of how far one drives the compression.

The interior operator: what happens at L_0

What happens at the boundary L_0 in the interior of a black hole is not fully open in FFGFT.

The T-field at its minimum. $\tilde{T}_{\text{min}} = \xi \cdot t_P > 0$ — a finite, positive, non-singular value.

Maximum winding density. One winding quantum per L_0^4 spacetime volume — analogous to the lower bound in a lattice with finite volume. This is the densest stable configuration on the ξ ladder (level $N = 0$). Below this level the state space ends.

Lagrangian stiffness. The term $-\frac{1}{2}m_T^2(\partial m)^2$ with $m_T = \xi/\ell$ (Dok. 201) prevents ∂m from diverging — algebraic property of the field equations, not an external force.

Topological bounce condition. No stable configurations below $N = 0 \Rightarrow$ no states the system can transition into. This is a **state space boundary**, not a force-counterforce mechanism.

Time in the interior. $\tilde{T} \to L_0/c$: local time quantum at its minimum. Time does not “freeze” ($\tilde{T} = 0$ would be required), but the local time structure reaches its smallest possible value. The modified Hawking spectrum arises precisely at this boundary.

Formal criterion for the transition operator. An interior operator would need to describe the mapping of an incoming winding configuration (n_i) onto the extremal configuration at L_0 — under preservation of topological charge, with maximum winding density. The T^4 algebra suggests this operator is **unitary**, but commutes non-trivially with the thermodynamic limit. That is the open research question — precisely stated, not a fundamental gap.

[Interior of a Black Hole in FFGFT] The interior is not unknown territory, but an extreme state:

- $\tilde{T}_{\text{min}} = \xi \cdot t_P > 0$ — finite, not singular
- Maximum winding density: $1/L_0^4$ per spacetime volume
- Lagrangian stiffness: ∂m does not diverge
- State space boundary at $N = 0$
- Unitary transition operator (details open)

Fundamental information is not the description of the totality

The fundamental information is the **class** of configurations (topological charge), not their **instance** (quantum state):

- Class: “There is an electron” (winding numbers defined)
- Instance: “The electron is at position x with spin $+\frac{1}{2}$” (quantum state)

FFGFT preserves the class. The instance can become thermodynamically irreversible.

.4 Black Holes as Inversion — Not as Destruction

The topological structure remains, inverted

A black hole is not a destruction state, but an **inversion** of normal matter structure: winding configurations maximally compressed, T-field at minimum L_0/c. What fell in as an electron remains topologically classifiable as an electron — in an extreme field regime.

Irreversibility is thermodynamic, not ontological

Irreversibility and information preservation operate on different levels:

- **Thermodynamic:** Directed, irreversible, entropy increases.
- **Topological:** Winding numbers **topologically preserved**. No destruction operator acts on topological invariants.

The Hawking paradox arises from confusing these two levels.

.5 The Radiation Mechanism: Node Tunnelling and Modified Spectrum

The preceding sections clarify what happens to the *information*. This section clarifies the *mechanism* of the radiation itself — why a black hole emits at all — and derives the FFGFT-specific spectrum modification from first principles.

Temperature and mechanism

The Hawking temperature remains **unchanged** in FFGFT (Dok. 201, §7.4):

$$T_H = \frac{\hbar\kappa}{2\pi k_B} = \frac{\hbar c^3}{8\pi G M k_B}, \qquad \kappa = \frac{c^4}{4GM}. \tag{3}$$

The surface gravity κ is geometrically fixed and is not modified by the T^4 structure. What changes is the **emission mechanism**: instead of particle-antiparticle pair production in a continuous vacuum, FFGFT emits **incoherent node excitations** that **tunnel** through the phase barrier at the horizon (Dok. 201, §7.4; Dok. 049). This is the FFGFT translation of the Parikh-Wilczek tunnelling picture: the quanta are winding excitations of the T-field, not modes of a continuous field.

Numerically (solar mass, $M = M_\odot$): $T_H \approx 6.17 \times 10^{-8}$ K; Schwarzschild radius $r_s \approx 2.95$ km. Since $T_H \propto 1/M$, evaporation accelerates as the mass decreases.

Entropy as node count

The Bekenstein-Hawking entropy is, in FFGFT, the **configuration entropy of the nodes**:

$$S = \frac{A}{4\ell_P^2} = N_{\text{node}} \cdot \ln 2, \qquad N_{\text{node}} \propto \frac{A}{\xi^2}. \tag{4}$$

Each node carries exactly one bit ($\ln 2$). For $M = M_\odot$: $S/k_B \approx 1.05 \times 10^{77}$, so $N_{\text{node}} \approx 1.5 \times 10^{77}$. This reproduces the area law exactly — no free adjustment.

Derivation of the spectrum modification from lattice dispersion

The central FFGFT prediction — a spectrum deviating from a pure black body — follows not from an ansatz but from the **minimal length** $L_0 = \xi \cdot \ell_P$. A minimal length implies a discrete lattice structure, and a lattice possesses a modified dispersion relation (analogous to phonon dispersion in a solid, Debye theory):

$$E(k) = \frac{2\hbar c}{L_0}\sin\left(\frac{kL_0}{2}\right) = \hbar ck\left[1 - \frac{(kL_0)^2}{24} + \mathcal{O}\left((kL_0)^4\right)\right]. \tag{5}$$

The leading correction is therefore:

$$\frac{\Delta E}{E} = -\frac{(kL_0)^2}{24} = -\frac{1}{24}\left(\frac{E}{E_{\text{max}}}\right)^2, \qquad E_{\text{max}} = \frac{\hbar c}{L_0} = \frac{E_P}{\xi}. \tag{6}$$

The cutoff $E_{\text{max}} = E_P/\xi \approx 7500 \cdot E_P \approx 9.2 \times 10^{22}$ GeV lies above the Planck energy. In units of the Hawking temperature ($E = x k_B T_H$):

$$\frac{\Delta E}{E} = -\frac{x^2}{24}\left(\frac{k_B T_H}{E_{\text{max}}}\right)^2. \tag{7}$$

Scale and falsifiability — an honest assessment

This derivation has the correct limiting behaviour, but its practical consequence must be **stated honestly**: for real black holes the modification is unmeasurably small.

For a stellar black hole ($M = M_\odot$) the typical quantum energy $k_B T_H$ is tiny against E_{max}:

$$\frac{k_B T_H}{E_{\text{max}}} \approx 5.8 \times 10^{-44} \quad \Rightarrow \quad \left.\frac{\Delta E}{E}\right|_{x=10} \approx -1.4 \times 10^{-86}. \tag{8}$$

The modification becomes significant only when $k_B T_H$ approaches E_{max} — that is, when the black hole has evaporated down to the mass scale

$$M_{\text{crit}} = \frac{c^2 L_0}{8\pi G} \approx 1.15 \times 10^{-13}\ \text{kg}. \tag{9}$$

This is precisely the **remnant scale**. The spectrum modification therefore arises exactly at the L_0 interface and at the evaporation endpoint — consistent with the interior-operator section.

[Correction of a naive ansatz] A naive modulation ansatz of the form $\exp\left(-\xi x^2/D_f\right)$ yields artificially large deviations ($\sim 0.4\%$) that physically **do not exist**. The correct deviation, derived from the lattice dispersion, scales as $(k_B T_H/E_{\text{max}})^2$ and is $\sim 10^{-86}$ for real black holes. A theory claiming a measurable 0.4% deviation would already be falsified. The correct FFGFT prediction is the much weaker but consistent statement: the spectrum is modified only at the evaporation endpoint.

Evaporation endpoint: remnant rather than destruction

As the black hole evaporates, M shrinks and with it the core radius. Approaching M_{crit}, the system reaches the state-space boundary $N = 0$ (maximal winding density $1/L_0^4$): there are no denser stable configurations into which it could transition. The result is a **stable remnant** (Dok. 201, §7.5):

- finite size $\sim L_0$,

- finite temperature,
- topological charge preserved in the remnant node state.

The original fundamental information thus remains encoded **on this side** — in the final stable node state. There is no transfer to an "other side" and no destruction. The thermodynamic time direction of evaporation remains irreversible; the topological preservation is unaffected by it.

.6 The Compactified Time Dimension

Time is not continuous — it is compactified

In FFGFT, the time dimension on T^4 is **compactified** — the winding number n_4 is a topological invariant like the others.

Important: Compactification means **periodicity of field configurations** in the time direction — not reversibility of the thermodynamic time. A macroscopic process remains irreversible. The T^4 topology is time-direction-neutral; the thermodynamic time direction is emergent, not fundamental.

What compactification changes

The statement "information is forever lost in the past" presupposes an absolute ontological past boundary. In a compactified time dimension, this absoluteness is not self-evident: the topological field structure is not referenced to the observer's time direction. n_4 remains topologically preserved.

Our experienced time is unambiguously directed

The experienced time direction is a thermodynamic phenomenon (entropy increase), not a geometric primitive. In FFGFT, the time direction is emergent, not fundamental — compactification operates at the topological level and does not mystically overload the theory.

.7 What FFGFT Concretely Claims — and What It Does Not

1. **No classical singularity:** $L_0 = \xi \cdot \ell_P$ as absolute lower density bound.
2. **Topological fundamental information (charge) preserved:** Winding numbers as topological invariants, algebraically protected by $\tilde{T} \cdot m = 1$ (Dok. 078).
3. **Modified Hawking spectrum:** Deviation from the black-body spectrum derived from the lattice dispersion, scaling as $(k_B T_H / E_{\text{max}})^2$. Unmeasurably small for real black holes ($\sim 10^{-86}$); significant only at the evaporation endpoint (remnant scale). Falsifiable in principle, but not in practice for astrophysical objects.
4. **Gravitational wave echoes:** After mergers as a further test.

Not claimed: Complete state reconstruction; carrier recovery; fully specified transition operator near $N = 0$; suspension of thermodynamic irreversibility.

Precise FFGFT Statement

FFGFT claims: the **topological fundamental charge** — the winding numbers of the infalling matter — is **topologically preserved**. It is not reconstructible by an external observer (epistemic access problem), but it continues to exist as a geometric property of the T^4 field.
If it were truly destroyed, the Hawking spectrum would be exactly thermal. The modified spectrum derived from the lattice dispersion is the difference in principle — though for real black holes unmeasurably small and significant only at the evaporation endpoint.

.8 Connection to the ξ Recursion

The ξ ladder defines which winding configurations on T^4 are stable. Black holes represent configurations at the lower end ($N = 0$) — within the ξ structure, not outside it. The "inverted" configuration of a black hole is also part of the ξ ladder and topologically well-determined.

Summary

Question	FFGFT Answer
Why does the paradox arise?	Hawking presupposes information as epistemic and time as continuous — neither fundamental in FFGFT
What is fundamental information?	Topological charge: winding numbers $(n_\theta, n_\varphi, n_3, n_4)$ on T^4 — not bits
What is not information?	Our knowledge of the state; carrier identity; complete state description
Is information lost?	Topological charge: no (topologically preserved). Detail state: access problem
Empirical grounding	Modified Hawking spectrum (lattice dispersion); difference in principle, practically unmeasurable for real black holes
Algebraic protection (Dok. 078)	$m \to \infty$ implies $\tilde{T} \to 0$ only asymptotically — singularity algebraically unreachable
What happens at L_0?	$\tilde{T}_{\text{min}} > 0$ · Max. winding density · Lagrangian stiffness · State space boundary $N = 0$
Is there a singularity?	No — L_0 as lower bound; state space ends at $N = 0$
Is the process irreversible?	Thermodynamically yes. Topologically: fundamental structure remains
Time compactification	Periodicity of field configurations, not reversal of thermodynamic time
Falsifiable tests	Hawking spectrum (modification only at evaporation endpoint); gravitational wave echoes
What remains open?	Transition operator $(n_i) \to$ extremal configuration at L_0 — unitary, details open

Addendum P35: ξ^N Triviality (7 September 2026)

Expressions of the form "$X = \xi^N$" are intrinsically meaningless on their own (P35, Doc. 190-Archive-2). Full justification: addendum P35 in Doc. 182. For the scale tables and ξ-powers in this document: only exponents whose form is derived geometrically forward are non-trivial. Retrospectively fitted exponents are to be classified as **[S]**.

Addendum R92(ii): Schwarzschild Radius and L_0 — Identity, not a Test (7 September 2026)

The old edition of Doc. 190 (23 August 2026, now Archive-2) stated under Precision 4: an object with $M_0 = \xi m_P/2$ would have Schwarzschild radius exactly L_0.

This is an identity, not a physical test. From $r_s = 2GM/c^2$ and $\ell_P = \sqrt{\hbar G/c^3}$ it follows that $Gm_P/c^2 = \ell_P$ (by definition), hence:

$$r_s(xm_P/2) = \frac{2G \cdot xm_P/2}{c^2} = x \cdot \frac{Gm_P}{c^2} = x\ell_P$$

for *every* $x \in \mathbb{R}^+$. With $x = \xi$ it follows that $r_s(\xi m_P/2) = \xi\ell_P = L_0$ trivially — the equation tests neither ξ nor the exponent in $L_0 = \xi\ell_P$. It is dimensional bookkeeping, not a substantive finding.

The classification as a "consistency check" in the old Doc. 190 was too strong. The relation $L_0 = \xi\ell_P$ is established by independent arguments **[K]** (Docs. 180, 327); this identity contributes nothing to that.